Case Study

Driving Factors for the Adoption of Digital Twin Technology Implementation for Construction Project Performance in Nigeria

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Abstract: The adoption of digital twin technology (DTT) for enhancing construction project performance, sustainability, and safety of construction workers, among others, is common practice in developed countries. Meanwhile, construction projects in developing nations are suffering from poor outcomes, which could benefit from DTT. Therefore, this research investigates the drivers of the adoption of DTT implementation in the Nigerian construction industry. Close-ended questionnaires were administered to digitally inclined professionals using purposive and snowballing techniques to elicit necessary information on the drivers of DTT implementation. These respondents include architects, engineers, quantity surveyors, and builders in the study area. It was found that technological advancement/trend, reliable data storage, safety, availability of software, customer satisfaction, and accessibility were the top-ranked drivers for implementing DTT. A Shapiro-Wilk test was conducted to know whether the data are normally distributed or not, which led to the use of the Kruskal-Wallis H-test. The Kruskal-Wallis test revealed that government policies have diverging views among the professionals, whereas other factors have converging opinions among the respondents. Factor analysis was conducted to group the key drivers for the implementation of DTT into innovation, operation, quality, performance, and policy drivers. The findings of this study will provide a reference point for researchers and construction organizations on the driving force that brings about the implementation of DTT in a business context and the construction industry. This study realizes that one of the key drivers of the adoption of this technology is a desire for innovation and technological advancement that can be achieved when the technology performs to the expectation of clients, organizations, and the architecture, engineering, and construction (AEC) industry. The study also suggests ways of implementing DTT in the construction industry as well. **DOI: 10.1061/JCEMD4.COENG-13659.** *© 2023 American Society of Civil Engineers.*

Practical Applications: A digital twin (DT) is a virtual replica or representation of a physical object or product. This research proposes practical implications of DT technology adoption in a developing country with a focus on major drivers for enhancing the uptake of the technology in the construction industry. This will give more insight to stakeholders on major drivers to improve its implementation in the construction and other sectors of the country as well as other developing countries. Technological advancement in different countries as shown in the study will expose professionals to the benefits of digital technologies as well as keep the professionals and other stakeholders abreast of the potential benefits. This can be realized when the government is at the frontline in the usage of DT for public construction projects. In addition, there is a need for policy formulation for its implementation in the construction industry to ensure projects are sustainable and delivered to the satisfaction of concerned stakeholders.

Author keywords: Construction industry; Digital twin (DT); Professionals; Pull factors; Technology; Construction performance.

Introduction

In general, the construction industry contributes around 7%–10% to the gross domestic product (GDP) of different countries, especially developing nations (Oladinrin et al. 2012; Saleh and Al-Swidi 2019). This sector provides employment, promotes growth and development, and is intertwined with other industries (Dixit et al. 2017). Meanwhile, the construction industry faces many issues, which include low productivity, inadequate communication, poor technological advancement, poor project performance, fragmentation, low innovation, and so on (Farmer 2016), despite several technologies available to improve productivity and performance in the industry. Besides, the construction industry has been reported to be slow at embracing innovative technologies compared with other sectors (Morledge 2011).

Through the fourth industrial revolution (4IR), the construction industry is experiencing fast-paced growth in different stages of building, such as design and engineering, construction, operation and maintenance, and demolition, in developed climes (Ebekozien and Aigbavboa 2021). However, its adoption is slow and the use of 4IR technologies is at a nascent stage in the architecture, engineering, and construction (AEC) industry in developing countries. The 4IR technologies such as augmented reality/virtual reality (AR/VR), cyber-physical systems, and DTs have brought increased

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Note. This manuscript was submitted on February 15, 2023; approved on September 5, 2023; published online on October 31, 2023. Discussion period open until March 31, 2024; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Construction Engineering and Management***, © ASCE, ISSN 0733-9364.

efficiency, productivity, and performance of work in various sectors including construction (Mihai et al. 2021). This has triggered the need for increased awareness and adoption of these technologies in construction due to the benefits derived. These benefits include the safety of workers, cost efficiency, timely delivery of projects, improved health, life cycle management of systems, construction waste management, and productivity improvement (Akanmu et al. 2021).

In particular, digital twin is not a new concept; it was introduced by Micheal Grieves in 2003 by presenting a virtual prototype of a physical product (Batty 2018; Grieves 2014). It is referred to as a two-way interaction between the physical entity and a virtual or digital replica for full integration (Alam and El Saddik 2017). There are numerous misconceptions about the nomenclatures of DT, such as digital models, digital shadows, and digital twins (Sepasgozar 2021). A digital model is just a digital representation of a physical system with no interaction, whereas a digital shadow is the one-way interaction of a digital replica and a physical entity. DT works when there is full integration with several enabling technologies such as building information modeling (BIM), wireless sensor networks (WSN), data analytics, and machine learning. Data analytics, artificial intelligence (AI), and machine learning help to enhance the capabilities of the DT in anomaly detection, optimization, prediction, and knowing the health status of equipment or systems (Boje et al. 2020).

Many researchers have examined DTs in various sectors like manufacturing, health, aeronautics and aviation, education, automotive, and so on (Tao et al. 2019a; Tuegel et al. 2011). In the construction industry, Opoku et al. (2022) carried out a systematic review of the drivers of DT adoption. It was revealed that developed countries like the US, Australia, the UK, and Italy were the countries advancing its usage compared with developing countries. Other researchers also examined DT application in construction, generally about workforce safety through reviews (Opoku et al. 2021; Hou et al. 2021; Ozturk 2021; Deng et al. 2021). Little or no attention has been paid to the empirical study of the drivers of the adoption of DT in the construction industry, especially in developing countries like Nigeria. This study will conduct empirical research on the drivers of DT implementation in developing countries based on the perception of professionals.

This research will give insight to practitioners, clients, government, educators, and other stakeholders concerning the pull factors to be considered for implementing DT in the Nigerian construction industry. It will help stakeholders to better understand the DT concept and propel readiness to embrace this technology. The rest of this study is structured as follows: section "Literature Review" reviews related and previous work on DT and the drivers. Section "Research Methods" discusses the research approach or methodology adopted in this study, and section "Analysis and Review" reveals the discussion of the survey conducted in relation to previous findings and the implications. The last section provides the conclusion, recommendations, and areas for further studies.

Literature Review

Introduction

This section contains a review of the driving factors that make DT to be adopted in any stages of built assets, i.e., design, construction, operation and maintenance, and demolition stage of built assets. These drivers of DT that can enhance implementation in the construction industry are synthesized logically.

Drivers of Digital Twin Technology for Construction Project Performance

A DT has three major components, which are a physical entity, connected data, and a virtual entity. Most previous research on DTs was largely based on reviews. This section reviews what has been stated by other researchers in the extant literature. However, this research conducted an empirical validation based on the opinions of professionals. Some of these factors are discussed in the following sections. The summary of the literature review was shown in Table 1.

Accessibility and Flexibility

Digital technologies have come to disrupt traditional ways of doing things in the construction industry. The issue of accessibility and flexibility should not be left out (Del Giudice et al. 2021). Fachrunnisa et al. (2020) noted that for digital transformation to be implemented in firms or organizations, there must be flexibility, agility, and adaptability of its use by workers. Also, there should be accessibility to data and information for use by organizations at any time. This information must be restricted from public use, i.e., data privacy should be guaranteed. Professionals and organizations should have equal access and opportunities to any digital technologies made available to improve the productivity and performance of construction or built assets.

Trust Level

The application of DTs in small-medium enterprises (SMEs) and large organizations requires trust. DTs will be implemented when they can demonstrate the separation of data and secure data. Herein, this includes the separation of nonvolatile and therefore non-real-time-dependent data such as ground plan, layout, purpose of the production, machine, parts list, qualification of employees, shifts, and so on. Volatile time-dependent data such as production flow, movements of employees, machine assignment, transport routes, and others come to bear (Uhlemann et al. 2017). When the trust level about digital security and privacy is minimal, it hampers the propensity of users of DTs in organizations. Government and large establishments would not want to adopt DTs if there is a high level of risk and public display of private or personal data. Also, there should be a high level of trust in the enumerated capabilities of DTs for employers to adopt (OECD 2016).

Government Policies

To ensure that the benefits of new technology reach every layer of society with enforced ethical consideration, the formulation of policies and regulations is required. There should be a major effort toward the democratization of technology. The different governmental bodies should initiate feasibility studies for utilizing DTs within their sectors. In particular, investigating different scenarios on how they can profit from projecting insights obtained from relevant predictive twins into their governmental responsibilities is important (Leng et al. 2019).

Policies implemented by the government do drive organizations to change the use of digital technologies. The components for creating DTs are very expensive, so there should be an arrangement for the government and organizations to implement these technologies for more productivity. The government can subsidies these technologies and give a chance for installment payments to the companies. This should be cut across every sector in the nation (Oke and Arowoiya 2021a).

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Table 1. Drivers of the adoption of DT technology for construction project performance

Code	Variables	A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z
D1	Technological advancement/trend	—	—	—	—	—	—	X	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
D2 and D5	Reliable data storage and reliability	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
D4	Availability of software	—	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—	—	—	—	—	—	—
D6	Customer/client satisfaction	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—
D7 and D17	Accessibility and flexibility	X	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—
D8 and D3	Quality and safety	—	—	—	—	X	X	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—	—
D9 and D12	Client and industry demand	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—	—	—	X	—	—	—	—
D10	User-friendliness	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—	—	—	—	—	—	—	X
D11	Drive for innovation	—	—	X	X	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
D13	Government policies	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	X	—	—
D14	Organizational acceptance	—	X	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
D15	Better management and performance	—	—	—	—	—	—	—	X	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	—
D16	Basic operation skills	—	—	—	—	—	—	—	—	—	—	—	—	X	X	—	—	—	—	—	—	—	—	—	—
D18	Trust level	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	X	—	—	—	—	—	—	—	—

Note: A = Del Giudice et al. (2021); B = Zou et al. (2017); C = Brem and Voigt (2019); D = De Jong and Marsili (2016); E = Raz et al. (2017); F = Tah and Carr (2020); G = Wang et al. (2014); H = Jo et al. (2018); J = Knapp et al. (2017); K = Wang et al. (2018); L = Jara et al. (2014); M = Yahya and Shafie (2018); N = Andrews et al. (2018); P = Oke and Arowoiya (2021b); Q = Uhlemann et al. (2017); R = OECD (2016); S = Kan and Anumba (2019); T = Fachrunnisa et al. (2020); U = Williams and Naumann (2011); V = Yu et al. (2020); W = Leng et al. (2019); X = Oke and Arowoiya (2021a); Y = Opoku et al. (2021); and Z = Oladinrin et al. (2023).

Friendliness

Being user-friendly is a valuable concept and an overworked phrase that has lost much of its meaning. User-friendly is not merely the addition of high-tech hardware such as a mouse, icons, or three-dimensional graphics. This refer to ease at which end users can access, navigate different softwares used in creating digital twin. For example, digital twin cloud services can aid in simulation and model which could learn by themselves. This give increase insights, easy-to-understand data and user friendly dashboards that can be utilized by teams in business or organization without the need of data experts (Oladinrin et al. 2023). We infer that user-friendly suggests an entity that is warm and comforting to the one who uses it (Jara et al. 2014).

Organizational Acceptance

In the last century, there was intense competition to enhance information and communication technology (ICT), which caused rapid changes and development of technologies. As a result, many organizations had to adjust and adapt to the changes in ICT. At present, ICT is being used as a tool to advance development in numerous fields (Zou et al. 2017). A DT is referred to as an innovatively new technology for executives and employees. Therefore, it is necessary to understand the benefits of the technology for the organizations to which they belong.

Drive for Innovation

Firms are trying to maximize their innovation efforts by efficiently serving their new or existing customers with new innovative or modified products and services, adding value in the customer's eyes (Brem and Voigt 2019). There was a belief that the main source of innovation was large enterprises. The increase in firm size leads to higher innovation capabilities; thus, large firms can make greater innovation developments than small ones. The ICT industry is characterized by continuous technological change, high research and development (R&D) expenditures, and innovation. In service sectors, innovation heavily depends on the skills and knowledge of employees, which are oriented to fit specific customers' wants by developing innovation in close interaction with clients (De Jong and Marsili 2016).

Quality and Safety

In any business organization or construction industry, profit and delivery of quality products are the ultimate goals. Profit may be maximized by cutting costs for the same selling price per unit. If it is a monopolistic business, without giving much importance to cost-reduction programs, the price may be fixed suitably to earn sufficient profit. To survive in a competitive business environment, goods and services produced by a firm should have the minimum required quality. Extra quality means extra cost, so the level of quality should be decided by other factors such that the product is well absorbed in the market. Quality is a measure of how closely a good or service conforms to a specified standard that can be achieved by a digital replica of the product (Raz et al. 2017).

Quality standards may be any combination of attributes and variables of the product being manufactured. The attributes will include performance, reliability, appearance, commitment to delivery time, and so on, and the variables may be some measurement variables like length, width, height, diameter, surface finish, and so on (Tah and Carr 2020). Quality begins with the design of a product to the customer's specifications. It further involves the established measurement standards, the use of proper material, the selection of a suitable manufacturing process, and so on. Quality is a relative term generally used concerning the end use of the product.

Concerning safety, the adoption of the DT in the construction process will facilitate a safer environment and improve efficiency by communicating real-time information to workers and visualizing happenings on the construction site and built environment (Kan and Anumba 2019).

Technological Advancement/Trends

Technology advancement is generally regarded as a catalyst for national development because it offers, among other things, the necessary support for change in all major sectors of the economy, most especially in the industrial sectors. Therefore, it is unarguably the prime source of change of innovations and adaptations required for improving production methods needed to propel growth and development. Technological advancement, which is required to acquire, select, or utilize source technology for company competitiveness, becomes the competitive edge and key capability to gain distinctiveness (Wang et al. 2014). Technological results and commercialization are the outcomes of technological development capability. There is a need to identify the capability required for successful technological development performance and technology commercialization for companies.

Better Management and Performance

The concept of performance is a reference both in theoretical approaches and practice. The field of economic performance includes various terms, the most important of which include competitiveness, productivity, profitability, and business growth. DT implementation can better the project management process and enhance effective stakeholder collaboration during the construction process. Activities in construction are properly managed with the use of DT applications in the construction industry (Opoku et al. 2021). DTs have shown considerable potential in different fields. However, the development of DTs is still in its infancy. No universal definition, implementation framework, and protocol are available. Therefore, it can be found that there lack of comprehensive and in-depth analysis of DTs from the perspective of concepts, technologies, and industrial applications (Jo et al. 2018).

Reliable Data Storage and Reliability

With the rapid development of embedded technology, embedded systems have widely infiltrated science, engineering, military technology, and other areas of daily life. Every field of study wants the reliability of the data stored. This has been a widely concerning issue, which affects product performance, stability, and security. Therefore, the reliability of embedded systems technology for data storage has become a new hotspot and may affect the future direction of the development of embedded systems (Knapp et al. 2017). Reliability refers to when a data storage system ensures data integrity and effectiveness. This is when data kept are not aborted, damaged, or lost. The reliability of data storage is affected by many factors such as power failure, memory failure, system instability, human error, and so on. They are a direct result of data storage reliability problems. Therefore, the reliability of data storage is a matter of hardware, storage technology, operating systems, and other areas of the comprehensive study (Wang et al. 2018).

Client and Industry Demand

The demand for DT technology is being felt across all sectors. From manufacturers who are already using DT to simulate their processes to aging hospitals that need to understand their assets so they can make informed decisions, and infrastructure projects, including critical assets such as pavements and rail, all are starting to see the benefits of combining building information modeling (BIM) with sensor technology (Yu et al. 2020). For business owners who want to take their business to the next level, the adoption of DT technology is essential (Jara et al. 2014).

Availability of Software

The development phase includes building a toolkit that can be used to create a unique DT for various assets. Toolkit development is the hardest part of the DT's life cycle. It is typically done in R&D by highly specialized data scientists, machine learning specialists, and more. The software development work involves gathering information about the physical characteristics of the asset, the operating environment, and any conditions under which the asset is operated. In addition, the automatically collected data from the asset are required to develop the required machine learning methodologies. The result is a mathematical model that can be used to model the behavior and performance of an asset (Yahya and Shafie 2018).

Basic Operation Skills

Technical skills are necessary for workers to use these technologies in construction or building assets. Workers and professionals must know how to effectively use these technologies in the organization (Andrews et al. 2018). Firms should be ready to train their workers with innovative technologies to improve the performance of projects. This training can also be incorporated into tertiary institutions to enhance students ability to meet work demands in the industry (Oke and Arowoiya 2021b).

Customer Satisfaction

Users and organizations must be satisfied and get value from any emerging technologies such as the DT they are using to improve their business. If there is a low level of satisfaction, it will surely affect the level of demand by managers of firms. Customer needs keep changing over time but emerging technologies should be able to meet the multidimensional needs of clients in construction. With the capabilities of DTs to perform diagnostics, prognostics, both predictive and prescriptive of any system accurately, it will propel more of its usage in the design, construction, operation, and maintenance of built assets (Williams and Naumann 2011).

Research Methods

A quantitative research method was adopted to investigate the drivers of DTs using a survey technique. Questionnaires were used to elicit information from construction professionals in Lagos state, Nigeria. Lagos state was chosen based on the level of growth of digital technologies among construction professionals in the industry compared with other states in Nigeria, the ever-increasing level of urbanization in the state, and the volume of construction projects being executed (Fagbenle et al. 2011).

Population

The study population was composed of architects, engineers, builders, and quantity surveyors. The process of selecting these professionals was through purposive and snowballing techniques, which means that not all professionals in Lagos state were included (Kothari 2014). Purposive sampling was based on the judgment of the researcher. Professionals with industrial technology skills were

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purposively selected (10 participants), who later identified the unknown respondents in the state, i.e., snowballing.

Data Collection

The research adopted questionnaires as the instrument for primary data collection targeted at addressing the research objective. The questions were asked on a five-point scale with 5 = Very high, 4 = High, 3 = Average, 2 = Low, and 1 = Very low. The Likert scale was chosen for ease and uniformity of responses. The Likert scale provides a more versatile and user-friendly means to address the drivers of DTs. The professionals were asked to rank how important the factors or variables can influence DT adoption.

The questionnaire designed for this research was such that the first section dwelt on the background information of the respondents, and the other section focused on the drivers of DT implementation on construction projects. The drivers used in the second section of the questionnaire were obtained from extant literature on drivers of DTs identified in previous studies. The questionnaire was administered to 369 construction professionals in the target area of study. The construction professionals work in contracting firms, consultancy firms, and government agencies that play important roles in building construction in Lagos state. Of the 369 copies of the questionnaire administered, 178 were found suitable for use, which represents a 48.2% response rate (Fosnacht et al. 2017).

Data Analysis

The following statistical techniques were used in analyzing the data: frequency, percentile, mean item score (MIS); standard deviation (SD), factor analysis, Shapiro-Wilk test, and Kruskal-Wallis H-test.

Frequency and percentiles were adopted to analyze the demographic information of the respondents. MIS was used for analyzing and ranking the opinions of the respondents on the drivers. SD was employed where two or more variables had the same MIS. An item with a lower SD was ranked higher than another variable with a mean score but a high SD. The Kruskal-Wallis H-test was used to test the significant difference in the opinion of four categories of respondents.

The Shapiro-Wilk test was used to test the normality of the distribution. This is used to know whether the factors are normally distributed or not (Corder and Foreman 2014). The result showed that the factors are not normally distributed, which led to the use of the Kruskal-Wallis test. Factor analysis was used to group variables into clusters for ease of discussion. The sample-to-item ratio in the study satisfied the minimum requirement of 5:1 for conducting factor analysis (Pallant 2020). Cronbach's alpha test was used in this research to test the reliability and validity of items that constitute each factor formed in the factor analysis (Taber 2018). In addition, Horn parallel analysis was conducted to determine the most suitable number of components for the factor analysis (Pallant 2007).

Analysis and Result

Background Information of Respondents

This section discusses the features of the respondent. It shows their years of working experience, highest academic qualification, profession, professional qualification, membership status, and years of operation of the organization of the professionals.

In total, 369 respondents were sent questionnaires. After the distribution, 203 questionnaires were retrieved, and 178 were deemed fit for analysis because some forms were incomplete. This shows a response rate of 48.2%; hence, the findings are reliable.

It is evident from Fig. 1 that 43% of respondents are registered with the Quantity Surveyor Registered Board of Nigeria (QSRBN) and 22% are registered with the Council of Registered Builders of Nigeria (CORBON). In addition, 14% of respondents are Architects Registration Council of Nigeria (ARCON) members, 18% are professionals registered with the Council for the Regulation of Engineering in Nigeria (COREN), and the remaining 3% are others. Also, in Fig. 2, the proportion of quantity surveyors is 44.9%, and architects represent 14.6%. Builders account for 19.1% and engineers represent 21.3% of respondents.

Furthermore, of all respondents, 60.7% were B.Sc./B.Tech. holders, 23% were M.Sc./M.Tech. holders, 11.2% held a Higher National Diploma (HND), and 4.5% had attained a Ph.D., as indicated in Fig. 3. Also, 22.5% of the construction professionals had

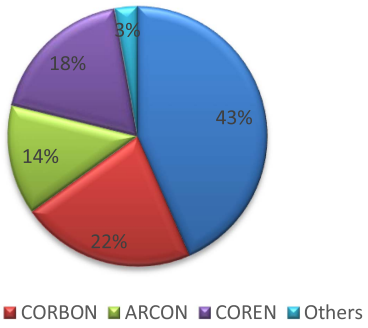


Fig. 1. Professional bodies for each profession.

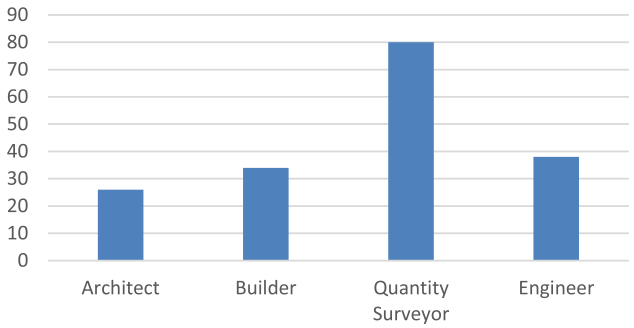


Fig. 2. Professions of respondents.

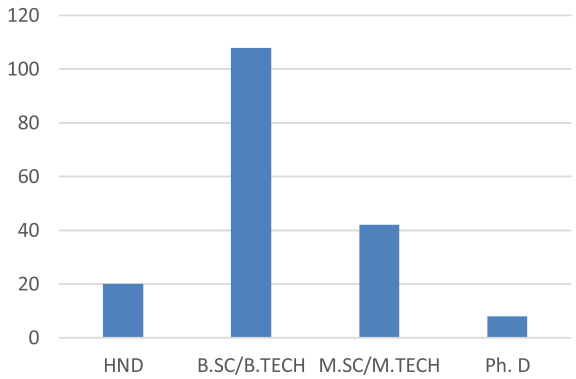


Fig. 3. Academic qualifications.

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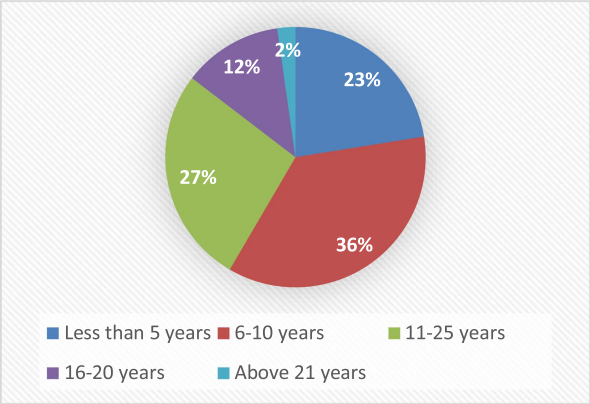


Fig. 4. Years of working experience.

fewer than 5 years of working experience, 36% had 6–10 years of working experience, 27.0% had 11–15 years of working experience,12% had 16–20 years of experience, and only 2% had 21 years or more of experience as revealed in Fig. 4. Moreover, 14.6% of organizations were established fewer than 5 years ago, 31.5% had 5–10 years of operation, 27.0% had 11–15 years of operation, 20.2% had operated for 16–20 years, and 6.7% had 21 years or more of operation. Based on this analysis, it can be concluded that the data provided by the respondents can be relied upon for the analysis.

Descriptive Statistics

Mean and Standard Deviation

Table 2 presents the mean score, SD, and rank of drivers of DT technology for construction based on the opinions of all the respondents, and the professional groups of the respondents as well. The result of the overall respondents' opinion reveals that technological advancement/trend (D1) was the first ranked driver, with a mean of 3.70, and the next variable ranked was reliable data storage (D2), with a 3.69 mean value. The third factor was safety (D3), with a mean of 3.69, and the fourth-ranked was the availability of software (D4), with a mean of 3.66. The fifth-ranked variable was reliability, with a mean of 3.63, and the least among the drivers was trust level (D18), with mean of 3.45. In comparison with previous studies, it has been revealed that safety risk management (Liu et al. 2021; Kaewunruen et al. 2021) was in line with this study. Other drivers identified from previous studies that do not corroborate or partially corroborate with the findings are improved product quality (Agostinelli et al. 2021), operation performance (Ozturk 2021; Lee and Lee 2021), personal satisfaction and others.

Inferential Statistics

Kruskal Wallis Test. Table 3 also presents the results of the Kruskal Wallis H-test used to identify the difference in the opinions of respondents on the drivers of the adoption of DT. Among the pull factors, the only factor having diverging opinions among professionals was government policies, whereas other variables have convergent views on other drivers, namely technological trend/advancement, industry demand, reliable data storage, safety, availability of software, reliability, customer satisfaction, client demand, organization acceptance, need for quality, user-friendliness, and so on. The observed significant difference in government policies could be linked to the size of the construction firms of the respondents. Construction professionals and stakeholders of large-sized enterprises may hold government policies in higher esteem than small and medium-sized organizations. Establishments at the survival stage are more concerned about attracting customers than following stipulated government policies that concern their business environment (Adeniyi et al. 2020).

Table 2. Drivers of digital twin technology for construction project performance

Code	Mean	SD	Rank
D1	3.70	0.845	1
D2	3.69	0.806	2
D3	3.69	0.912	3
D4	3.66	0.904	4
D5	3.63	0.817	5
D6	3.63	0.817	5
D7	3.62	0.846	7
D8	3.62	0.886	8
D9	3.62	0.898	9
D10	3.62	1.061	10
D11	3.61	0.778	11
D12	3.60	0.734	12
D13	3.58	0.877	13
D14	3.55	0.917	14
D15	3.54	0.813	15
D16	3.48	0.827	16
D17	3.47	0.813	17
D18	3.45	0.812	18

Horn Parallel Analysis. The minimal level of factor loading acceptance is 0.3 or 0.4, which serves as a rule of thumb to determine significance (Hair et al. 2010). The factor loading value for this study was set at 0.3 in the Statistical Package for the Social Sciences (SPSS) software version 28, and all the loadings were above the cut-off value, which was retained. To determine the number of factors to be extracted, horn parallel analysis was conducted after the actual eigenvalue was conducted for principal component analysis. Horn's parallel analysis is said to be the most accurate method for determining the number of factors to retain compared with Kaiser's criterion and scree test, which made us consider it in the study (Pallant 2007). Both Kaiser's criterion (eigenvalue exceeding 1) and the screen test revealed the presence of five components that need to be verified in parallel analysis. With the aid of Horn's parallel analysis as the most accurate method to determine the number of factors to be extracted (Pallant 2007), all five components were supported and accepted. The rule governing the parallel analysis is that the actual eigenvalue from principal component analysis (PCA) (SPSS output) must be greater than the criterion value from parallel analysis as indicated in Table 4.

Factor Analysis. The adequacy of the sampling approach was considered in terms of size of the set data and the extent of the population. According to Tabachnick and Fidell (2007), Bartlett's test of sphericity should be significant ($p < 0.05$) for factor analysis to be suitable. They also propounded that a Kaiser—Meyer—Olkin (KMO) index should range from 0 to 1, with 0.50 considered suitable for factor analysis. In this case, the KMO index is 0.786, and the test of sphericity is significant with a value of 0.000 (Table 5); therefore, factor analysis is suitable in both cases (Hadi et al. 2016).

Table 6 provides the initial matrix and extracted matrix of squared loading results for the drivers of DT for construction project performance. The Total, Amount of variance (%), and Cumulative amount (%) columns are initial matrices; the next three columns are the rotated matrix only when the eigenvalue is greater than 1. The total variance explained by each factor is listed in the Total subcolumn under the Initial eigenvalue column. In the third

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Table 3. Kruskal Wallis test on the drivers of the adoption of DTs in the construction industry

		Architect		Builder		Quantity surveyor		Engineer		Overall		Kruskal-Wallis	
Code	S-W	MIS	RK	MIS	RK	MIS	RK	MIS	RK	MIS	SD	Chi-square	Asymp. sig.
D1	0.000	4.08	1	3.29	15	3.83	3	3.53	6	3.70	0.845	7.257	0.064
D13	0.000	3.62	11	3.12	18	3.88	2	3.37	15	3.58	0.877	10.430	0.015*
D12	0.000	3.62	12	3.59	2	3.65	13	3.47	7	3.60	0.734	1.147	0.766
D4	0.000	3.62	13	3.53	4	3.78	8	3.58	4	3.66	0.904	0.793	0.851
D5	0.000	3.77	5	3.35	11	3.78	7	3.47	10	3.63	0.817	3.075	0.380
D6	0.000	3.54	14	3.35	9	3.75	10	3.68	1	3.63	0.817	3.082	0.379
D7	0.000	3.46	17	3.47	5	3.73	11	3.63	2	3.62	0.846	1.002	0.801
D8	0.000	3.62	10	3.35	10	3.80	6	3.47	9	3.62	0.886	3.937	0.268
D9	0.000	3.69	7	3.35	12	3.77	9	3.47	11	3.62	0.898	3.538	0.316
D10	0.000	3.85	4	3.29	16	3.83	4	3.32	17	3.62	1.061	6.296	0.098
D11	0.000	3.54	15	3.65	1	3.67	12	3.47	8	3.61	0.778	0.939	0.816
D3	0.000	3.69	6	3.29	17	3.90	1	3.58	5	3.69	0.912	5.646	0.130
D2	0.000	3.85	2	3.35	8	3.80	5	3.63	3	3.69	0.806	3.658	0.301
D14	0.000	3.85	3	3.29	14	3.62	15	3.42	12	3.55	0.917	4.190	0.242
D15	0.000	3.62	8	3.41	7	3.62	14	3.42	13	3.54	0.813	0.890	0.828
D16	0.000	3.23	18	3.47	6	3.63	16	3.37	16	3.48	0.827	2.746	0.432
D17	0.000	3.46	16	3.53	3	3.58	17	3.21	18	3.47	0.812	3.224	0.358
D18	0.000	3.62	9	3.29	13	3.48	18	3.42	14	3.45	0.812	0.870	0.833

Note: S-W = Shapiro-Wilk value; MIS = mean item score; RK = rank; SD = standard deviation; and Asymp. sig. = asymptotic significant values. *p-values less than 0.05.

Table 4. Comparison of eigenvalues from PCA and criterion values from Horn's parallel analysis

Component number	Actual eigenvalue from PCA	Criterion value from parallel analysis	Decision
1	5.826	1.273	Accepted
2	1.794	1.047	Accepted
3	1.392	0.871	Accepted
4	1.272	0.734	Accepted
5	1.106	0.612	Accepted

Table 5. KMO and Bartlett's test for drivers of digital twin technology for construction project performance

Test type	Result
Kaiser-Meyer-Olkin measure of sampling adequacy	0.785
Bartlett's test of sphericity	
Approximate chi-square	598.701
Degrees of freedom (DF)	178
Significance	0.000

column, the factors are arranged in descending order of variance explained. In the third column of Table 6, the first five items have a cumulative variance of 63.27%, which implies the remaining 13 items account for only 36.73% of the variance.

The Scree plot in Fig. 5 is a graphical representation of the eigenvalues on the *y*-axis against the factors represented as components on the *x*-axis of the plot. The scree plot for a factor analysis always displays a downward curve; a graphical explanation of the total variance as explained in Table 7.

As presented in Table 8, a model of five items was deemed adequate to represent the drivers of DT for construction project performance. Therefore, only five reduced components were obtained from this factor analysis. Table 8 presents the varimax rotation, and each variable weighs heavily on one of the factors, and loading on each factor exceeds 0.30. According to Yap (2013), it was recommended that it is better to interpret only factor loadings greater than 0.30. The high communalities are expressions that the extracted components represent the variables well.

Discussion of Findings

As presented in Table 6, a model of five items was deemed adequate to represent the drivers of DT for construction project performance. Therefore, only five reduced components were obtained from this factor analysis. It becomes highly imperative to name the five extracted components before a proper and concise interpretation can be made. From the literature, there exists little or no known principle for naming extracted components of factor analysis (Henson and Roberts 2006). The implied meaning of the preceding statement is that naming these factors could be subjective and based on the perception, background, and training/education of the viewer/researcher/analyst. After careful consideration of the variables grouped into each component, the following names were deemed appropriate: (1) innovation drivers, (2) operation drivers, (3) quality drivers, (4) performance drivers, and (5) policy drivers.

Innovation Drivers

Innovation drivers, which was the first component extracted, contains five items, and it depicts a driver that has to do with processes involved in organizing a construction project. Also, it refers to concept-driven factors that would enhance improvements in design, project planning, delivery, safety, and decision making. The five items under this category are as follows: drive for innovation, with a factor loading of 0.785; industry demand, with a factor loading of 0.734; accessibility, with a factor loading of 0.712; and availability of software, with a factor loading of 0.680, and safety, with a significance of 0.384. In reality, the drive for innovation is essential for adopting technology by clients, organizations, governments, and professionals (Adafin et al. 2022). Construction stakeholders without enthusiasm for technology may prefer the use of traditional approaches as long as they can achieve the project goals. The inertia to change construction stakeholders forms an impediment to the

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Table 6. Total variance explained of the drivers of digital twin technology for construction project performance

Component	Initial eigenvalues			Rotation sums of squared loadings		
	Total	Amount of variance (%)	Cumulative amount (%)	Total	Amount of variance (%)	Cumulative amount (%)
1	5.826	32.366	32.366	5.826	32.366	32.366
2	1.794	9.966	42.332	1.794	9.966	42.332
3	1.392	7.733	50.064	1.392	7.733	50.064
4	1.272	7.069	57.133	1.272	7.069	57.133
5	1.106	6.144	63.278	1.106	6.144	63.278
6	0.974	5.413	68.690	—	—	—
7	0.881	4.894	73.584	—	—	—
8	0.756	4.201	77.785	—	—	—
9	0.676	3.755	81.540	—	—	—
10	0.584	3.244	84.784	—	—	—
11	0.530	2.942	87.726	—	—	—
12	0.439	2.439	90.165	—	—	—
13	0.423	2.353	92.517	—	—	—
14	0.376	2.087	94.605	—	—	—
15	0.296	1.647	96.252	—	—	—
16	0.271	1.504	97.755	—	—	—
17	0.226	1.256	99.011	—	—	—
18	0.178	0.989	100.000	—	—	—

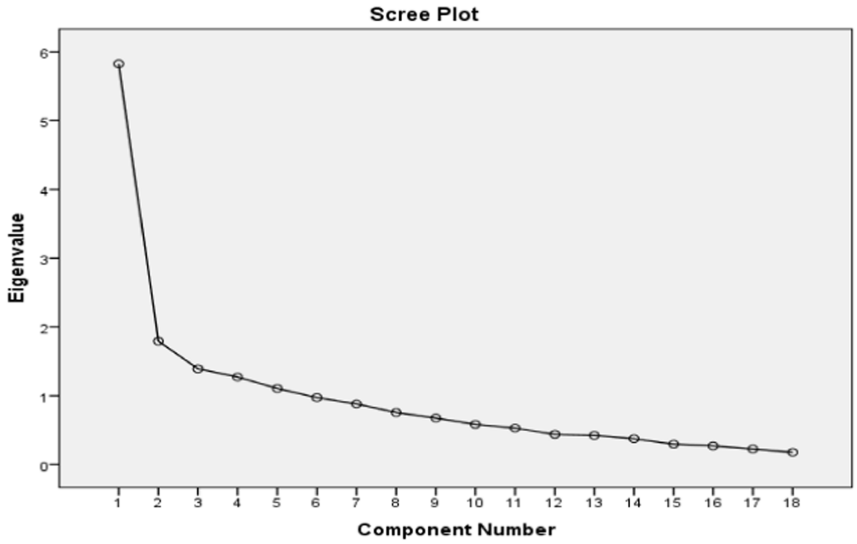


Fig. 5. Graphical representation of eigenvalues.

adoption of digital technology (Olawumi et al. 2018). This corresponds with the findings of Liu et al. (2021), who revealed that DTs can be adopted in innovative approaches in product development, construction process, and safety management in the construction industry.

In addition, the accessibility to real-time data of DTs is incomparable with BIM, which deals with static data, which is an indication of an upgrade in innovation. This is in line with Agostinelli et al. (2021), who noted that a DT of the building can enrich knowledge and data available through sensors and AI techniques. This means that a DT can monitor and develop self-learning and predictive capabilities for systems or subsystems in small or large projects. According to Opoku et al. (2021), the application of DTs ameliorates the duration of the overall design and construction process which is partly in line with this finding. It was emphasized that the optimization of information can assist in reducing the possibility of additional costs for the project. This implies proper optimization of information can minimize cost and time overrun in construction. Therefore, the availability of software and other technological devices for setting up DTs of buildings, systems, processes, products, and so on are essential for the wide adoption of DTs to achieve intended goals in the construction industry.

Operation Drivers

Operation driver is the name allotted to the second extracted component. As the name implies, the variables of operation related loaded strongly under the same factor and so have been grouped. It contains four items, which are flexibility, with a factor loading of 0.641; basic operation skills, with a factor loading of 0.615; trust level, with a factor loading of 0.599; and reliability, with a significance of 0.482. The flow of information and convenience for stakeholders' use is germane for its successful implementation. A DT must develop a platform for a user interface that can be used by the general public in a business context and organization. For example, when a DT is used in the area of energy management, the

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Table 7. Rotated component matrix of drivers of digital twin technology for construction project performance

Variables	Component 1	2	3	4	5
Drive for innovation	0.785	—	—	—	—
Industry demand	0.734	—	—	—	—
Accessibility	0.712	—	—	—	—
Availability of software	0.680	—	—	—	—
Safety	0.384	—	—	—	—
Government policies	—	0.820	—	—	—
Client demand	—	0.779	—	—	—
Customer satisfaction	—	0.741	—	—	—
Flexibility	—	—	0.756	—	—
Basic operation skills	—	—	0.724	—	—
Trust level	—	—	0.643	—	—
Reliability	—	—	0.482	—	—
Better management and performance	—	—	—	0.839	—
Reliable data storage	—	—	—	0.751	—
Technological advancement/trend	—	—	—	0.649	—
User-friendliness	—	—	—	—	0.826
Organizational acceptance	—	—	—	—	0.731
Need for quality	—	—	—	—	0.495

Note: Extraction method: principal component analysis; and rotation method: varimax with Kaiser normalization. Rotation converged in seven iterations.

operation becomes efficient, using prognostics, monitoring of real-time information, and so on with ease, which promotes its adoption (Opoku et al. 2022). Despite the flexibility and user-friendly interphase in digital technologies, there are still dichotomies in their trust level. It is sometimes believed that the adoption of technology may lead to the violation of privacy and confidential information of the organization, whereas others believe that end-to-end encryption of technology devices can be trusted to assuage, reduce, or mitigate any privacy infiltration (Patel and Patel 2020).

Quality Drivers

Quality driver is the name allotted to the third extracted component. As the name implies, the quality-related variables loaded strongly under the same factor and so have been grouped. It contains three items: user-friendliness, with a factor loading of 0.826; organizational acceptance, with a factor loading of 0.731, and need for quality, with a significance of 0.495. Technology and its devices minimize the difficulty of executing any given task and thus does DTs to the construction projects, the industry, and the entire ecosystem through replicating physical systems for ease of decision making (Ford and Wolf 2020). The user-friendliness and flexibility of digital technologies are motivators for their adoption in various industries (Gangwar et al. 2015).

In fact, organizational acceptance of technology is associated with the confidentiality of data from other competitors. In the last century, there was intense competition to enhance ICT, which caused rapid changes and development of the relevant technologies (Zou et al. 2017). As a result, many organizations had to adjust, adapt, and accept the changes brought by ICT because it is an indispensable tool for advancing development in the construction industry and other fields. Although the construction industry is often reported to lag in the adoption of technology among other sectors (Morledge 2011), top management officers in construction organizations must understand DTs' benefits for wide acceptance in the industry.

Performance Driver

Performance drivers are the fourth extracted component. It consists of three items, namely, better management and performance, with a factor loading of 0.839; reliable data storage, with a factor loading of 0.751 and technical advancement/trend, with a factor loading of 0.649. The application of DTs in the design, construction, or operation phases should enhance performance in terms of cost and time reduction (Opoku et al. 2021). This is an important aspect because digital technologies such as DTs could improve the time, cost, and safety performance of construction projects. Large construction projects consist of many interconnected parts, which may lead to cost and time overruns if not well-managed (Leung and Olomolaiye 2010). However, through DTs, areas where performance-related deficiencies that could lead to increased budgets and excessive time can be easily identified, and addressed accordingly (Kim and Kim 2020; Tao et al. 2019b).

Table 8. Reduced components for the drivers of digital twin technology for construction project performance

S. No.	Factor components	Variables	Factor loading	Alpha value
1	Innovation drivers	Drive for innovation	0.785	0.776
		Industry demand	0.734	
		Accessibility	0.712	
		Availability of software	0.680	
		Safety	0.384	
2	Policy drivers	Government policies	0.820	0.635
		Client demand	0.779	
		Customer satisfaction	0.741	
3	Operation drivers	Flexibility	0.756	0.673
		Basic operation skills	0.724	
		Trust level	0.643	
		Reliability	0.482	
4	Performance drivers	Better management and performance	0.839	0.733
		Reliable data storage	0.751	
		Technological advancement/trend	0.649	
5	Quality drivers	User-friendliness	0.826	0.633
		Organizational acceptance	0.731	
		Need for quality	0.495	

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On the other hand, having reliable data storage is crucial for construction projects in recent decades. Having reliable storage data storage for previous and ongoing projects can help to understand and solve similar problems on any future projects promptly (Meža et al. 2021). In fact, the data available could serve as a buffer for construction project designers and other construction project stakeholders to guide against some salient problems in the industry. In addition, the need for reliable storage data also relates to recent technical advancements/trends in the industry and other sectors, in which project actors want to make meaning out of data for decision making and overall performance of construction projects (Stojanovic et al. 2018; Lu et al. 2020). In addition, the advantage of DTs in providing real-time data on the project is an essential driver for construction projects' performance.

Policy Drivers

The fifth extracted component is policy drivers. It consists of three items, namely, government policies, with a factor loading of 0.820; client demand, with a factor loading of 0.779, and customer satisfaction, with a significance of 0.741, respectively. When there is legislation and policy on the usage of innovative technology, it will improve client demand, and its usage brings satisfaction (Agrawala 2019). The policy will enhance the implementation of DT usage even though clients are not willing to adopt these technologies. Interestingly, the government is the most influential client in the construction industry (Fox and Skitmore 2007). According to Leng et al. (2019), the benefits of the new technology should reach every layer of society and actors in construction projects, which can be spearheaded by the government. This implies that different governmental bodies should initiate feasibility studies for utilizing DTs within their sectors. In particular, they should investigate different scenarios on how they can profit from projecting insights obtained from relevant predictive twins into their governmental responsibilities. Prior studies have proved that government policies play a crucial role in adopting innovation in the construction industry (Hazarika and Zhang 2019).

On the other hand, enforcement of the policies with fines for defaulters has been an impetus through which industries in developing countries embraced any policies (Adeniyi et al. 2020). Although large enterprises working in developing nations are more proactive in welcoming government policies, micro, small, and medium-sized establishments are a bit laid back. Perhaps, large organizations know the implications of breaching government law and policies on their reputation. The use of DTs on construction projects could be linked with client demand because the adoption of the technology would translate into substantial financial outlay on the part of the organizations, which would be recouped from the project owner at the end (Kshetri 2021). Thus, a client with limited funds to execute a project may be reluctant to request the use of any digital technology on the project. However, a financially buoyant client may not demand the technology if they are deficient in the knowledge of the benefits of the project.

Conclusion, Recommendations, and Further Studies

Conclusion

A DT is the digital representation of a physical object with connected data. This real-time data helps in managing buildings, improving productivity, and decision making. A DT has various capabilities depending on the enabling technologies used in developing DT technology. Proper orientation and awareness of these levels or hierarchy of DTs and how it resolves issues and enhances satisfaction in the industry are critical in its adoption. This study examined the pull factors for using DT technology in the construction industry using descriptive and inferential statistics. The descriptive statistics revealed that technological advancement/trend, government policies, industry demand, availability of software, reliability, customer satisfaction, and client demand were ranked the topmost drivers of DT implementation. The results of the Kruskal Wallis H-test indicated converging views of the groups of professionals on 17 of the drivers, except for government policies. The divergent opinions of the respondents on government policies are linked to the size of the respondent's establishment.

Factor analysis was conducted to group the drivers of DTs into manageable sizes. The factor analysis was conducted to affirm the critical drivers, components, and horn parallel analysis to ascertain the number of factors to be extracted. The five components that the factors were grouped into are innovation, operation, quality, performance, and policy drivers. The findings of this study will provide a reference point for researchers and construction organizations on the driving forces that bring about the implementation of DTs in a business context and the construction industry. The study gave insight into the pull factors of DTs in the construction industry.

Recommendations

Some recommendations from the study to improve the adoption of DTs in the construction industry are as follows:

- Emphasis should be made by the appropriate bodies such as private clients, construction developers, professionals, and different stakeholders in the construction industry on the functionalities and benefits derived from using DTs in construction, built assets, and service firms, such as safety and reliability in disseminating stored information, which have been examined in this study, so that it will spur the acceptance of stakeholders.
- Firms and organizations must be ready to train their staff and workers on digital technology skills such as DTs because such basic operation skills have been stated as one of the drivers of DTs to be able to solve challenges faced in the construction industry.
- Government should enact a policy on the use of digital technologies such as BIM, which was enforced in the UK for public projects. This will facilitate the use of these technologies. Also, the purchase of these technologies should be subsidized for firms/organizations so clients and other stakeholders can demand these technologies.
- The use of technologies must be based on business value and technological drive. Organizations that adopt these technologies must be satisfied, which enhances the implementation. The choice of the capabilities of the DT should be based on the challenges faced by the organizations and their size. This will further enhance the productivity and performance of companies in executing their project.

Limitations and Areas of Further Study

The study was limited to Lagos state, Nigeria, but the findings can be applied to other states of the federation, and to developing countries where development is similar. The limitation of the study is that a qualitative study was not carried out to broaden the perception of professionals. A survey of construction professionals was mainly used to obtain opinions on drivers of DTs in this study. Although this approach can be considered suitable to achieve the aim of the study, there may be other drivers that were not captured in the research instrument. Therefore, future studies can explore other drivers of DTs via focus groups or Delphi surveys from

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experts to affirm this study in other developing climes. In addition, further studies can be carried out by examining the propelling force of using DTs for built assets from the user/clients' perspective or project managers. A case study of how DTs can bring value to built assets, i.e., asset management, fault detection and diagnosis, predictive maintenance, energy reduction, and predictive energy consumption, can be conducted in developing countries such as Nigeria.

Data Availability Statement

All data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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